

EEE6503 Spring 2026

Project #3

Goal(s)

Propose a long-tail visual recognition method for an image classification task.

Brief Summary

Visual recognition datasets at modern scale often contain highly uneven class distributions: only a few target categories make up a large portion of the training data, while many others have far fewer examples. Standard training procedures on such data tend to overfit to frequent classes, introducing bias and reducing performance on rare categories. To overcome this limitation, long-tail recognition approaches aim to enable models to extract class-discriminative representations even when the number of samples varies significantly across classes. This project investigates long-tail image classification, with experiments conducted on imbalanced versions of the CIFAR-100 dataset.

Instructions

- You should use the provided imbalanced CIFAR100 datasets **only**.
- You may split the training dataset into training/validation sets to set hyperparameters.
- Feel free to exploit various training techniques to boost the performance.
- We limit the maximum number of training epochs to **200** with a batch size fixed to **128**.
- You should use the provided test split **only** to evaluate your model at the final stage.
- DO NOT tune your model using the test dataset.
- Define your own validation set within the training split.

Submission

- **Due: PM 11:00, June 23rd (submit via LearnUS)**
- Include your report (*.pdf), code (*.ipynb), and model weights (*.pth)
 - *: [Group ID]

Grading Policy

- Report (30 points)
 - Format (5 points): Is your report well-formatted?
 - The report should include the following sections: introduction, proposed method, experimental results, and concluding remarks.
 - The maximum length of the report is 6 pages, excluding references. References are mandatory.
 - Use the CVPR author kit for LaTeX:
https://cvpr2022.thecvf.com/sites/default/files/2021-10/cvpr2022-author_kit-v1_1-1.zip
 - Technical Novelty (10 points): Is your proposed approach technically novel?
 - Analysis (5 points): Is your analysis valid?
 - Your experimental results should clearly demonstrate the effectiveness of your approach.
 - Make sure you include both quantitative and qualitative results.
 - Performance (10 points): Does your network balance well between efficiency and effectiveness?
 - You should report the results for 4 long-tail scenarios.
 - Your network will be evaluated based on accuracy, the number of parameters, and FLOPs.

Recommended references

We offer some long-tail recognition papers below.

- “Feature Transfer Learning for Face Recognition with Under-represented Data”, X. Yin et al., CVPR 2019 ([link](#))
- “Decoupling Representation and Classifier for Long-tail Recognition”, B. Kang et al., ICLR 2020 ([link](#))
- “Learning Imbalanced Datasets with Label-Distribution-Aware-Margin Loss”, K. Cao et al., NeurIPS 2019 ([link](#))
- “Long-tail Learning via Logit Adjustment”, A. Menon et al., ICLR 2021 ([link](#))

If you have any questions or concerns, feel free to contact TAs via email: shoon.lee@yonsei.ac.kr or geon.lee@yonsei.ac.kr